

6 Seaweeds are a group of organisms that carry out photosynthesis.

Identifying the types and proportions of chlorophyll pigments is important in the classification of seaweeds and biodiversity studies.

(a) State the meaning of the term **biodiversity**.

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(b) Explain the role of chlorophyll in the light-dependent reactions of photosynthesis.

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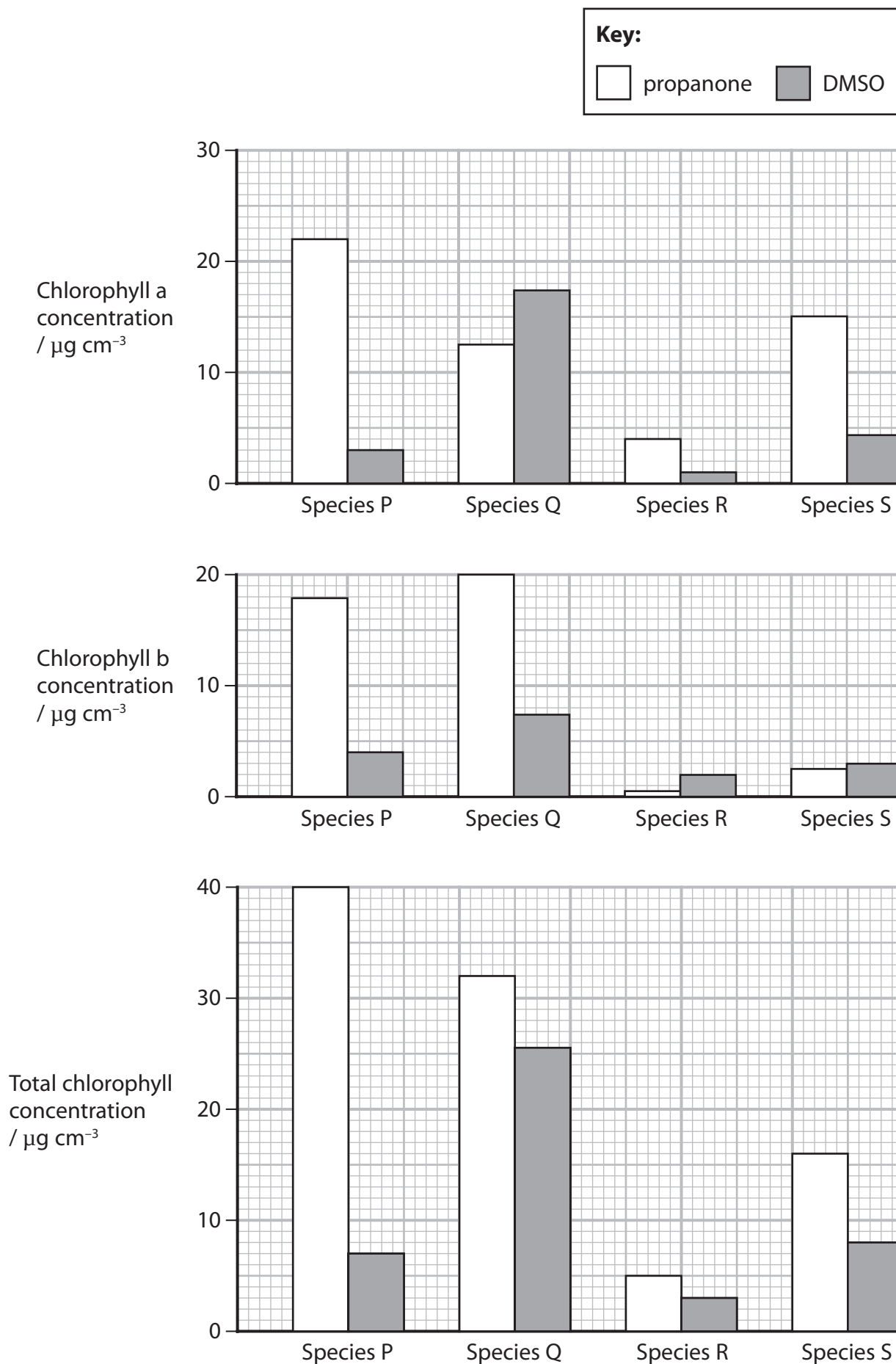
- (c) An investigation compared the effectiveness of two solvents, propanone and DMSO, in extracting photosynthetic pigments.

Four species of seaweed were collected from the Indian coast of Tamil Nadu and taken to the laboratory where the species were identified.

Each species of seaweed was split into two samples of equal mass.

The photosynthetic pigments were extracted from each sample using one of the two solvents and their concentrations determined.

The graphs show some of the results of this investigation.



- \*(i) Discuss the effectiveness of these extraction methods in the identification of these species.

Use the information in the graphs to support your answer.

(6)

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(ii) Suggest why different concentrations of chlorophyll were obtained using these two different solvents.

(2)

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**(Total for Question 6 = 12 marks)**

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